

INSTRUCTOR WALKTHROUGH

PROJECT 03 · BHĀRAT FOODS & BEVERAGES CO.

FMCG Sales & Distribution Analytics Q2 FY2026 Review

Grading guide covering Q1-Q5 (Channel Mix, Tier Expansion, Category Health, Distributor Health Score, ₹40 Cr Investment Strategy). Dual capstone — Q4 and Q5 are separate but linked deliverables.

Companion to	Bharat_Foods_FMCG_Case_Study_Instructions.pdf
Data file	Bharat_Foods_FMCG_Data.xlsx (9 sheets, ~76K rows)
Framework	PRISM — 5 questions across two capstones
Key difference	Dual capstone: Q4 = Distributor Health Score, Q5 = Investment Strategy
Confidentiality	NOT for student distribution

▲ Confidentiality

This walkthrough contains verified model answers and grading benchmarks. Do not distribute to students. Students receive the Solution Deck and Workbook only.

1. PRISM Applied to FMCG Distribution

FMCG distribution analytics differs from digital marketing (Project 1) and CRM/support (Project 2). The KPIs are primary sales, secondary sales, Sec/Primary ratio, return rate, and distributor health. Two decision layers: operational (distributor quality) and strategic (investment allocation). PRISM handles both.

P

Problem Definition

CMD's question: 'How do we deploy ₹40 Cr to maximize Q3 growth?' The real decisions: (1) Where are we growing and where are we stalling? (2) Which distributors are dragging the network? (3) Which category trends are real vs promo-propped? Not 'which city gets money' — but 'which combination of tier expansion, channel mix shift, and distributor upgrade delivers the best risk-adjusted return.'

R

Reframe into Data Questions

Five sub-questions: (1) GT+MT vs GT-only growth differential and per-distributor productivity. (2) Tier-1/2/3 growth rates and market penetration white space. (3) Category-level health: growth quality (organic vs promo-propped), return rates, scheme ROI. (4) Which distributors are underperforming — build a scoring model. (5) Given findings, how should ₹40 Cr be allocated for Q3?

I

Investigate the Data

Cross-sheet joins required: Primary_Sales → Distributors on distributor_code (adds city_tier, channel_type); Secondary_Sales → Distributors similarly; Returns → Primary_Sales for return rate. Date formats mixed across sheets. Health Foods scheme cost deeply buried in Trade_Schemes — easy to miss. **Most critical finding to catch in I: Health Foods scheme cost is 19.8% of sales, and return rate is 31.6% — without this, Q3 students recommend scaling a loss-making category.**

S

Solve from Multiple Angles

Six cuts: channel type × growth, city tier × growth × penetration, category × growth/return/scheme, distributor × sales/overdue/visit frequency, SKU × contribution, state × Sec/Primary ratio. **Health Foods 81.6% growth collapses to a weak business when return rate and scheme cost are netted: net sales ₹4.39 Cr, scheme ROI 3.46x. Tier 2+3 is 0.9% penetrated in a ₹5,321 Cr market. Both findings only emerge from decomposition.**

M

Measure Impact and Communicate

Dual deliverables: (1) Distributor Health Score — 0-100 scorecard with feature weights, validated at-risk list, and intervention tiers. (2) ₹40 Cr Investment Model — allocation by priority with quantified expected return. CMD sees the growth story. CFO sees the investment math. Sales Head sees the distributor action list.

◆ Teaching tip for Project 3

Students frequently conflate primary sales growth with business health. Push them hard on Health Foods: 'Is 81.6% growth actually good news?' The answer — 31.6% return rate, 19.8% scheme intensity, scheme ROI 3.46x — is a business that only grows when you spend heavily on promotions and then takes back a third of what you shipped. Q3 recommendation should be to pause Health Foods scaling, not accelerate it.

2. Question-by-Question Answer Key

► Q1 — Channel Mix Analysis (PRISM: S)

Headline:

GT+MT distributors grew +28.6% Apr→Sep vs GT-only +7.6%. GT+MT earns ₹16.6L avg per distributor vs GT-only ₹14.7L (+13%). With only 143 GT+MT distributors vs 657 GT-only, the network is structurally underinvested in Modern Trade reach.

Metric	GT-only	GT+MT	Implication
Distributor count	657	143	Modern Trade presence is minority of network
Avg sales/dist (₹L)	14.7	16.6	+13% per distributor productivity premium
Growth (Apr→Sep)	+7.6%	+28.6%	4× faster growth for GT+MT
% of total sales	80.2%	19.8%	Underweight given performance differential

◆ What strong students also compute

GT+MT share of total primary sales is only 19.8% despite growing 4× faster. If GT+MT distributors were 30% of the network (not 15%), projected incremental revenue = ~₹8-12 Cr annually. This is the investment case for Q5.

Answer Key · continued

► Q2 — City Tier Expansion (PRISM: S)

Headline:

Tier 2 growing +19.8%, Tier 3 growing +27.5% vs Tier 1 only +5.9%. But Bharat Foods penetration in Tier 2+3 is only 0.9% of a ₹5,321 Cr market. The whitespace is enormous.

City Tier	Growth	Current Sales (₹ Cr)	Penetration	Market Size (₹ Cr)
Tier 1	+5.9%	73.5	61.7%	28,926
Tier 2	+19.8%	40.1	32.2%	4,666
Tier 3	+27.5%	6.5	13.0%	655

★ The 0.9% Tier 2+3 penetration insight

Current Tier 2+3 sales = ₹46.6 Cr. Total Tier 2+3 market = ₹5,321 Cr. Penetration = 0.9%. Even achieving 2% penetration = +₹60 Cr revenue. The growth case for Tier 2+3 is structural, not promotional.

► Q3 — Category Health (PRISM: S)

Headline:

Health Foods 81.6% growth is promo-propped and returns-impacted. Beverages is declining −7.7% with weak secondary pull (Tamil Nadu Sec/Prim = 0.07). Biscuits and Snacks are the healthy volume drivers.

Category	Primary (₹ Cr)	Growth	Return Rate	Scheme Cost	Verdict
Biscuits	68.5	+11.8%	3.5%	Low	Strong — backbone of business
Snacks	32.0	+9.0%	7.4%	Low	Solid — grow with GT+MT
Health Foods	6.4	+81.6%	31.6%	19.8%	WARNING — pause scaling
Beverages	13.1	−7.7%	9.3%	Moderate	Fix pull before push

◆ Health Foods return rate math

Gross sales ₹6.42 Cr. Returns ₹2.03 Cr (31.6%). Net sales ₹4.39 Cr. Scheme cost ₹1.27 Cr. Scheme ROI = ₹4.39 Cr net / ₹1.27 Cr scheme = 3.46x. Against a 5x+ ROI benchmark for new category launches, this is below threshold. The growth is driven by pushing product through trade schemes, not by consumer pull.

Answer Key · Q4 & Q5 Capstones

► Q4 — Distributor Health Score (PRISM: S→M)

No single correct weighting. Grade the feature selection, logic, validation, and actionability of the output.

Feature	Source	Direction	Suggested Weight
Avg monthly sales	Primary_Sales	Lower = lower score	25-30%
Payment overdue days	Primary_Sales	Higher = lower score	20-25%
Last ASM visit (days ago)	Distributors	Higher = lower score	15-20%
Return rate	Returns	Higher = lower score	15-20%
Months active (6-month period)	Primary_Sales	Lower = lower score	10-15%
Tenure years	Distributors	Lower = lower score	5-10%

★ What top students also do

(1) Tier the at-risk list into three groups: Critical (<40), Watch (40-60), Healthy (60+). (2) Generate reason strings: 'Distributor X — not visited 87 days, avg overdue 18 days, return rate 22%.' (3) Recommend ASM prioritisation: Critical = visit within 7 days. (4) Compute what % of total revenue is at risk from Critical distributors.

► Q5 — ₹40 Cr Investment Strategy (PRISM: M)

The model answer. Students must ground Q5 in findings from Q1-Q3 — allocation logic must flow from evidence.

Bucket	₹ Cr	Q1-Q3 Grounding	Expected Return
New Tier 2 distributors	15	0.9% penetration in ₹5,321 Cr market	₹25-35 Cr revenue in 2 years
GT→GT+MT conversions	8	GT+MT growing 4× faster than GT-only	₹8-12 Cr incremental/year
Biscuits + Snacks deepening	8	Highest-volume categories, healthy returns	Volume + margin expansion
Beverages pull activation	5	–7.7% growth, weak Sec/Primary ratio	Arrest decline, rebuild sell-through
ASM + digital capability	4	28.1% not visited 60+ days = network bleed	Distributor retention

Bucket	₹ Cr	Q1-Q3 Grounding	Expected Return
TOTAL	40	All allocations data-grounded	~₹40-55 Cr incremental 2yr revenue

▲ What NOT to accept in Q5

Allocating significant spend to Health Foods scaling without caveating the return rate. The correct treatment: Health Foods gets zero or minimal incremental Q3 investment — instead, diagnose the return root cause (near expiry + slow moving = distribution mismatch or overstocking) and fix that first before scaling.

3. Expected Excel Artifacts

Join	Source Sheets	Key Column	Purpose
Primary_Sales ← Distributors	Primary_Sales, Distributors	distributor_code	Adds city_tier, channel_type to every transaction
Secondary_Sales ← Distributors	Secondary_Sales, Distributors	distributor_code	Required for Sec/Primary ratio by tier/channel
Returns ← Primary_Sales	Returns, Primary_Sales	distributor_code	Compute return rate per distributor and category
Trade_Schemes → Primary_Sales	Trade_Schemes, Primary_Sales	category/sku	Scheme cost as % of sales — critical for Health Foods verdict

Pivot	Rows	Values	Purpose
Primary by channel × month	month_year; channel_type	SUM billed_value_inr	Q1 growth differential
Primary by tier × month	month_year; city_tier	SUM billed_value_inr	Q2 tier trend
Sec/Primary by distributor	distributor_code	SEC/PRIM ratio	Q2 network health
Category growth + return rate	category	growth; return_rate	Q3 scorecard
Health Foods scheme intensity	category	scheme_cost/primary	Q3 promo-propped signal
Overdue by distributor	distributor_code	avg overdue	Q4 health score input
Returns by reason	return_reason	COUNT/SUM	Q3 return root cause

◆ Critical formula for scheme ROI

=SUMIFS(scheme_cost, category, "Health Foods") / (SUMIFS(billed_value, category, "Health Foods") - SUMIFS(return_value, category, "Health Foods")) → Result: 3.46x. Benchmark: ≥5x for a new category launch. This single computation changes the Q3 investment recommendation from 'scale Health Foods' to 'pause and fix.' Students who skip the scheme join miss this entirely.

4. Hidden Insights Cheatsheet

1. Health Foods 81.6% growth collapses to 3.46x scheme ROI after netting returns

Most students see 81.6% growth and recommend scaling. The join: Trade_Schemes → Primary_Sales → Returns. Net sales = ₹4.39 Cr. Scheme cost = ₹1.27 Cr. ROI = 3.46x vs 5x benchmark. The category only grows when you push via promotions — consumer pull is absent.

2. Tier 2+3 is 0.9% penetrated in a ₹5,321 Cr market

City_Demographics sheet. Bharat Foods current Tier 2+3 sales = ₹46.6 Cr. Total market = ₹5,321 Cr. Even 1.5% penetration = +₹33 Cr incremental. The growth case is structural.

3. GT+MT distributors grow 4× faster with 13% higher per-distributor productivity

+28.6% vs +7.6%. ₹16.6L vs ₹14.7L avg. But GT+MT is only 143 of 800 distributors. The network is underinvested in Modern Trade reach.

4. Beverages Sec/Primary in Tamil Nadu = 0.07 — secondary pull nearly absent

Secondary pull measures actual sell-through at retail. A ratio of 0.07 means for every ₹100 pushed to distributors, only ₹7 is selling through to consumers. Pipeline buildup without consumer pull = stale stock and future returns.

5. 28.1% of distributors not visited in 60+ days — network bleed at scale

Distributors not visited regularly drop out of active selling. $28.1\% \times 800 = 225$ distributors at risk. Average sales per distributor = ₹15L. Potential annual revenue bleed = ~₹34 Cr.

6. 33.4% of distributors generating <₹10L — low-productivity drag on network

These distributors consume ASM time and credit facilities proportionate to larger ones, but deliver far less revenue. Identifying them enables focused upgrade or restructuring.

◆ Grading note

5-6 insights → 90th percentile (full marks Diagnostic Depth). 3-4 → 70th. 1-2 → 50th. Consistent with Projects 1 and 2.

5. Common Student Mistakes

1. Recommending to scale Health Foods without examining return rates

Feedback: 'Your Q5 allocated ₹10 Cr to Health Foods expansion based on 81.6% growth. Health Foods return rate is 31.6% — nearly one-third of shipments come back. Net sales are ₹4.39 Cr, and scheme cost is ₹1.27 Cr (19.8% of gross). Scheme ROI of 3.46x is below the 5x benchmark for new category launches. Q5 recommendation should pause Health Foods scaling and diagnose returns first.'

2. Reporting city tier sales totals without computing penetration

Feedback: 'You correctly identified Tier 2 growing +19.8%. To complete the story: Bharat Foods current Tier 2+3 sales are ₹46.6 Cr against a ₹5,321 Cr total market — 0.9% penetration. The growth rate is the signal; the penetration gap is the opportunity. That comparison is what justifies the Tier 2 investment in Q5.'

3. Not computing Sec/Primary ratio

Feedback: 'Your Q2 analysed primary sales by tier but didn't compute Sec/Primary ratios. Secondary/Primary = actual sell-through vs push. Beverages in Tamil Nadu has a Sec/Primary ratio of 0.07 — meaning only ₹7 in ₹100 is selling through to consumers. That's a distribution pull problem, not a category problem. It changes the intervention.'

4. Q4 distributor score with only 1-2 features

Feedback: 'Your distributor health score used only payment overdue days. A robust score needs 5-6 features: sales volume, overdue days, visit recency, return rate, months active, tenure. More importantly, the score needs to produce an actionable at-risk list — not just a number. Who are the top 50 at-risk distributors, and what does each risk string look like?'

5. Q5 investment allocation without grounding in Q1-Q3 data

Feedback: 'Your ₹40 Cr allocation split equally across four buckets without connecting to findings. The rubric's cross-section integration dimension (15%) specifically tests whether Q5 flows from Q1-Q4. GT+MT growing 4x faster justifies the GT-to-GT+MT conversion bucket. 0.9% Tier 2+3 penetration justifies the tier expansion bucket.'

6. Grading Rubric

Dimension	Weight
Data Quality & Joins	10%
Diagnostic Depth	20%
Business Acumen	20%
Q4 Health Score Quality	15%
Q5 Investment Alignment	15%
Cross-Section Integration	10%
Stakeholder Communication	10%

Detailed benchmarks

Data Quality & Joins (10%)

Bracket	What it looks like
90% — Excellent	Primary→Distributor join done. Scheme→Primary join done. Returns rate computed by category. All cross-sheet work documented.
70% — Solid	Most joins correct. Minor gaps in scheme attribution.
50% — Functional	Single-sheet analysis. No joins.

Diagnostic Depth (20%)

Bracket	What it looks like
90% — Excellent	5-6 hidden insights found. Health Foods scheme ROI computed. Beverages Sec/Primary state-level computed. GT+MT premium quantified.
70% — Solid	3-4 insights. Growth rates correct. ROI or Sec/Primary missing.
50% — Functional	0-2 insights. Surface-level aggregations only.

Business Acumen (20%)

Bracket	What it looks like
90% — Excellent	Health Foods scale-down justified with numbers. Tier 2 case built from penetration math. Distributor visit gap quantified as revenue risk.
70% — Solid	Correct directional recommendations. Rupee impact missing or estimated.
50% — Functional	Qualitative only — no quantified implications.

Q4 Health Score Quality (15%)

Bracket	What it looks like
90% — Excellent	5-6 features. Validated. At-risk list with reason strings. Intervention tiers (Critical/Watch/Healthy). LTV at risk computed.
70% — Solid	3-4 features. Partial validation. List without reason strings.
50% — Functional	1-2 features. No validation. No at-risk output.

Q5 Investment Alignment (15%)

Bracket	What it looks like
90% — Excellent	All allocations flow from Q1-Q4 findings explicitly. Each bucket has quantified expected return. Constraints acknowledged.
70% — Solid	Sensible split. Partial grounding in data.
50% — Functional	Equal split or arbitrary. No data grounding.

Cross-Section Integration (10%)

Bracket	What it looks like
90% — Excellent	Q4 risk list explicitly feeds Q5 distributor upgrade bucket. Q3 Health Foods finding explicitly pauses that category in Q5.
70% — Solid	Q4 and Q5 are connected but loosely.
50% — Functional	Q4 and Q5 treated as independent exercises.

Stakeholder Communication (10%)

Bracket	What it looks like
90% — Excellent	CMD/CFO/Sales Head each get tiered summaries. Exec summary survives 4-minute test.
70% — Solid	One stakeholder framing. Generic language.
50% — Functional	Data dump. No executive synthesis.

7. Sample Feedback + Educator Notes

Scenario	Suggested feedback
Strong (90%+)	<i>'Outstanding. Health Foods scheme ROI correctly computed at 3.46x against the 5x benchmark, and Q5 correctly pauses scaling pending return root cause diagnosis. Tier 2 penetration math (0.9% of ₹5,321 Cr market) is precisely the right framing for the investment case. Q4 health score uses 6 features with reason strings for the Critical tier. One push to reach 95%: quantify the revenue at risk from the Critical distributor list explicitly.'</i>
Solid (70-79%)	<i>'Strong diagnostic work — GT+MT premium and tier growth correctly identified. Two gaps: (1) Health Foods verdict is incomplete — you flagged the return rate but didn't compute scheme ROI (3.46x vs 5x benchmark). That changes the Q5 recommendation. (2) Q4 score uses 3 features — add return rate and ASM visit recency for a robust score.'</i>
Below threshold	<i>'Your Q5 recommendation allocated ₹12 Cr to Health Foods expansion based on +81.6% growth. This is the most common and most costly mistake in this case. Join Returns → Primary_Sales by category: Health Foods return rate = 31.6%. Then join Trade_Schemes: scheme cost = 19.8% of sales. Net scheme ROI = 3.46x — below the 5x launch benchmark. Q5 investment should pause Health Foods and prioritise Tier 2 expansion instead.'</i>

★ Closing note for educators

Project 3 introduces FMCG distribution complexity: multi-sheet joins, scheme economics, and a dual capstone that tests both operational scoring (Q4) and strategic allocation (Q5). The cross-section integration dimension (10%) specifically rewards students who close the loop — using Q4 risk outputs in the Q5 distributor upgrade bucket. Spend 10 minutes in class on this connection. It's the difference between a good analytics student and a business-ready one.

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